

Project Charter

HEALTHCARE ANALYTICS CLOUD FOUNDATION

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DOCUMENT REVISION LOG

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1. Project Overview

HealthFirst Diagnostics (fictional) is establishing a secure and scalable cloud foundation to support the Healthcare Operations Analytics Platform that will be used for analyzing laboratory operational data, with Microsoft Fabric implementation, producing Power BI dashboards.

This project delivers the foundational cloud environment required to support the future deployment and operation of the Healthcare Operations Analytics Platform, including the governance, identity management, resource organization, and cost controls consistent with enterprise best practices.

2. Objectives

Provide a secure, well-governed cloud foundation capable of supporting Healthcare Operations Analytics Platform, including:

- Establishing secure cloud identity management
- Creating a governed Azure resource structure
- Enabling secure storage of analytics data
- Establishing a cloud environment capable of supporting future BI and analytics workloads
- Implementing basic automation
- Providing a maintainable development environment
- Minimizing operational cost during the initial implementation phase

3. Project Role

Project Role: Solution Architect

Responsible for designing and implementing the foundational Azure cloud environment required to support the Healthcare Operations Analytics Platform. Responsibilities include solution design, cloud governance, identity management, resource organization, cost and lifecycle planning, documentation, and implementation of foundational automation.

4. Scope

In Scope:

- Identity management
- Resource organization
- Cloud storage
- Networking foundation
- Provisioning the Microsoft Entra ID Federated Trust framework to support future automated continuous deployment (CI/CD) pipelines from the GitHub Repository control plane
- Automated Identity Validation and Trust Handshake (OIDC) via GitHub Actions
- Serverless Azure Data Lake Storage Gen2 (ADLS Gen2) with a custom Medallion container directory layout
- Governance documentation

Out of Scope:

- Kubernetes
- Production deployment
- Disaster recovery
- High availability
- CI/CD pipelines for physical infrastructure deployment
- Enterprise monitoring
- Production data migration
- Azure DevOps
- Terraform
- Large-scale infrastructure
- Fully automated continuous deployment pipelines (CI/CD) via GitHub Actions

5. Deliverables

Deliverable	Description
Microsoft Entra Application Registration	Registered application identity for future services
Azure Resource Group	Development resource organization
Storage Account	Secure storage for analytics datasets
Virtual Network	Private networking foundation
ADLS Gen2 three-tier container structure (Bronze/Silver/Gold)	Medallion Data Architecture in the Data Lake storage container for organizing raw, cleansed, and curated data
Automated Identity Validation and Trust Handshake (OIDC) workflow file (.github/workflows/validate-azure-identity.yml)	GitHub Actions automated workflow that connects to Microsoft Entra ID using OpenID Connect (OIDC) to execute a passwordless handshake, validating repository authentication without using static secrets
GitHub Repository	Source control and project documentation
Cloud Architecture Diagram	High-level solution architecture
Engineering Decisions Document	Design rationale
Cost & Lifecycle Management Strategy	Operational Governance

6. Assumptions

- Azure subscription available
- Development environment only
- Sample data used during implementation
- Single Azure region
- Existing Microsoft Entra tenant available
- Future analytics solutions will build upon this environment

7. Constraints

- Solution must remain extensible for future analytics initiatives
- Project duration limited to initial foundation implementation
- Production healthcare data is outside project scope

8. Success Criteria

- Identity services are operational
- Resource organization follows defined standards
- Cloud storage is operational
- Automation workflow executes successfully
- Architecture documentation is complete
- Governance documentation is approved
- Environment supports future analytics deployment
- All deployed resources follow project naming conventions and tagging standard

9. Risks

Risk	Mitigation
Uncontrolled cloud costs	Resource sizing, lifecycle management
Resource sprawl	Naming conventions and governance
Unauthorized access	Microsoft Entra ID and least privilege
Scope expansion	Defined project scope
Inconsistent resource naming	Naming standards document